V HACK 2026

CASE STUDY BOOKLET

Contents

CASE STUDY 1

Predictive Maintenance for SME Resilience

CASE STUDY 2

Digital Trust – Real-Time Fraud Shield for the Unbanked

CASE STUDY 3

First Responder of the Future – Decentralised Swarm Intelligence

CASE STUDY 4

The Inclusive Citizen – Multilingual AI for Public Services

CASE STUDY 1:

Predictive Maintenance for SME Resilience

Track: Machine Learning (Time-Series / Remaining Useful Life Estimation)

Primary Goal: SDG 9: Industry, Innovation, and Infrastructure (Target 9.4)

1. Real-World Context

Small and Medium Enterprises (SMEs) are the backbone of the ASEAN economy, yet they often operate with aging industrial machinery and thin profit margins. Unlike large conglomerates, these factories cannot afford "Smart Factory" overhauls. A single motor failure in a rural food processing plant can halt production for weeks, leading to massive financial losses and resource waste.

2. Problem Statement

Current maintenance in ASEAN SMEs is largely reactive (fixing after failure) or preventative (replacing parts too early). Both are inefficient. There is a critical need for an AI-driven system that can analyze sensor data,such as vibration, temperature, and load to predict the Remaining Useful Life (RUL) of machinery, enabling proactive planning and reducing downtime.

3. Technical Challenge & Sub-tasks

- **Temporal Feature Engineering**: Process high-frequency multivariate time-series data to extract features that represent machine degradation.
- **RUL Regression Modeling**: Develop a robust model to estimate the precise number of cycles or hours remaining before a functional failure.
- **Anomaly Change-Point Detection**: Implement logic to identify the exact moment a machine transitions from a "Healthy" state to an "Impaired" state.
- **Health Dashboard Visualization**: Design a user-friendly interface for factory managers to visualize machine health and schedule maintenance efficiently.

4. Technical Feasibility & Constraints

- **Noisy Data Handling**: Models must demonstrate the ability to handle sensor noise and missing data points common in real-world industrial settings.
- **Model Interpretability**: The AI must provide actionable insights (e.g., why the RUL is decreasing) to gain trust from non-technical factory operators.
- **Scalability**: The solution should be modular, allowing it to be applied to different types of machinery with minimal retraining.

5. Recommended Data Sources & Toolkits

- **Suggested Datasets**: NASA CMAPSS.
- **Suggested Frameworks**: Scikit-Learn, TensorFlow/PyTorch, Darts, sktime, NeuralProphet, or Prophet.
- **Visualization Tools**: Streamlit or Dash.

6. Expected Deliverables

- **Predictive Model**: A validated model with clear performance metrics (e.g., RMSE or MAE).
- **Functional Dashboard**: A prototype showing real-time machine status and "Time-to-Failure" countdowns.

7. Reference Section

[1] "NASA Turbofan Jet Engine Data Set,"www.kaggle.com. *https://www.kaggle.com/datasets/behrad3d/nasa-cmaps*

[2] "Welcome To Zscaler Directory Authentication," Cambridge.org, 2026. *https://www.cambridge.org/core/journals/ai-edam/article/remaining-useful-life-prediction-methods-of-equipment-components-based-on-deep-learning-for-sustainable-manufacturing-a-literature-review/C3FFF4402D1EF1EC9BDD1F5C84198BC1* (accessed Feb. 11, 2026).

[3] "UCI Machine Learning Repository," archive.ics.uci.edu. *https://archive.ics.uci.edu/dataset/791/metropt+3+dataset*

[4] "UCI Machine Learning Repository," archive.ics.uci.edu. *https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset*

CASE STUDY 2:

Digital Trust – Real-Time Fraud Shield for the Unbanked

Track: Machine Learning (Fraud & Anomaly Detection)

Primary Goal: SDG 8: Decent Work and Economic Growth (Target 8.10)

1. Real-World Context

Digital payment adoption in ASEAN is skyrocketing, particularly through "Super Apps." However, millions of new users are "unbanked" or have low digital literacy. For a gig worker in the Philippines or a rural merchant in Thailand, losing their digital wallet balance to a fraudulent transaction is catastrophic and destroys trust in the digital economy.

2. Problem Statement

Conventional rule-based fraud detection systems are often inadequate in identifying sophisticated or evolving fraud patterns. There is an urgent need for AI models that can analyze transaction behavior, detect anomalies in real-time, and prevent fraudulent activities without disrupting genuine transactions for legitimate users.

3. Technical Challenge & Sub-tasks

- **Behavioral Profiling**: Analyze user patterns such as transaction frequency, amount, location, and time to build a "normal" behavior baseline.
- **Real-Time Anomaly Scoring**: Develop a model that can score a transaction in milliseconds to "Approve," "Flag," or "Block."
- **Imbalanced Class Handling**: Implement techniques (like SMOTE or focal loss) to handle datasets where fraudulent transactions are extremely rare.
- **Contextual Data Integration**: Incorporate non-transactional data (e.g., IP address reputation or device fingerprints) to enhance detection accuracy.

4. Technical Feasibility & Constraints

- **Low Latency**: The system must operate in real-time. High-latency models that slow down the checkout process are not viable.
- **False Positive Control**: The model must minimize "False Positives" to avoid blocking legitimate transactions, which frustrates users.
- **Privacy-First**: The solution must demonstrate ethical handling of sensitive financial data (Data Integrity).

5. Recommended Data Sources & Toolkits

- **Suggested Datasets**: IEEE-CIS Fraud Detection dataset, Credit Card Fraud Detection (Kaggle).
- **Suggested Frameworks**: XGBoost / LightGBM, PyOD.
- **Deployment Tools**: FastAPI.

6. Expected Deliverables

- **Fraud Detection Engine**: A trained model optimized for high precision and recall in detecting anomalies.
- **Risk API Prototype**: A mock-up showing how a digital wallet would receive and act on a risk score.

7. Reference Section

[1] "IEEE-CIS Fraud Detection," kaggle.com. *https://www.kaggle.com/c/ieee-fraud-detection*

[2] M. Harris, O. Bennett, C. Lewis, E. Jackson, and M. Yusuff, "AI-Powered Fraud Detection in Digital Payments," ResearchGate, Apr. 2025. *https://www.researchgate.net/publication/400508360_AI-Powered_Fraud_Detection_in_Digital_Payments*

[3] E. Lopez-Rojas, "Synthetic Financial Datasets For Fraud Detection," www.kaggle.com, 2017. *https://www.kaggle.com/datasets/ealaxi/paysim1*

[4] "Anti Money Laundering Transaction Data (SAML-D)," www.kaggle.com. *https://www.kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml*

CASE STUDY 3:

First Responder of the Future – Decentralised Swarm Intelligence

Track: Agentic AI (Decentralised Swarm Intelligence)

Primary Goal: SDG 9: Industry, Innovation, and Infrastructure (Target 9.1 & 9.5)
SDG 3: Good Health and Well-being (Target 3.d)

1. Real-World Context

The ASEAN region, situated on the Pacific Ring of Fire, is highly vulnerable to super typhoons and earthquakes. In the critical first 72 hours, terrestrial communication (cell towers and internet) often suffers catastrophic failure, creating a "communication blackout." Standard AI models that depend on the cloud become useless. This case study challenges you to build a "self-healing" rescue swarm that operates as a collective brain at the edge, ensuring aid reaches survivors even when the world is offline.

2. Problem Statement

Current rescue models rely on high-bandwidth cloud connectivity for processing and coordination. When infrastructure collapses, these centralized systems fail. There is an urgent need for an Autonomous Command Agent that can operate at the edge to orchestrate a fleet of rescue drones. This system must be capable of mapping disaster zones, detecting survivors using on-device AI, and efficiently managing drone resources through a standardized Model Context Protocol (MCP) to ensure aid reaches survivors even without human pilots or cloud access.

3. Technical Challenge & Sub-tasks

- **Autonomous Mission Planning:** Develop a central "Command Agent" capable of decomposing a high-level goal (e.g., "Scan the South-East quadrant for thermal signatures") into specific, sequenced tool calls.
- **MCP Tool Integration**: Build an MCP server that exposes drone functions,such as move_to(x,y), get_battery_status(), and thermal_scan(),as standardized tools for the agent to use.
- **Strategic Resource Management:** Implement reasoning logic so the agent manages multiple drones effectively, ensuring search coverage is optimized and drones are recalled for "charging" (simulated) before battery failure.
- **Real-time Tool Discovery**: The agent must not have hard-coded drone IDs. It must use the MCP discovery mechanism to "see" which drones are active on the network and adapt its plan based on the available fleet.

4. Technical Feasibility & Constraints

- **Simulation-Only**: This is a software simulation. No physical hardware is required. Participants can use a 2D grid or a simple Python-based environment to represent the disaster zone.
- **Mandatory Protocol**: All communication between the Agent (the LLM) and the Drones (the code) **must** be handled via the **Model Context Protocol (MCP)**. Hard-coding drone movements is prohibited.
- **Agentic Reasoning**: The system must demonstrate "Chain-of-Thought" reasoning, where the agent explains its logic (e.g., *"Drone A has 20% battery, so I am assigning the closer Sector 1 to it while sending Drone B to the further Sector 5"*) before executing the tools.

5. Recommended Data Sources & Toolkits

- **Suggested Simulation Environments:** Mesa or a FastAPI mock environment.
- **Suggested Agent Frameworks:** LangChain, Microsoft AutoGen, or Vercel AI SDK.
- **Suggested Connector:** MCP Python SDK.

6. Expected Deliverables

- **The Orchestrator:** A functional AI agent capable of managing a simulated fleet of at least 3-5 drones.
- **MCP Server Implementation:** The code for the MCP server that exposes the drone "tools" to the agent.
- **Mission Log:** A demonstration showing the agent's step-by-step reasoning and its successful completion of a search-and-rescue mission.

7. Reference Section

The Protocol

[1] "Introduction - Model Context Protocol," Modelcontextprotocol.io, 2025. *https://modelcontextprotocol.io/introduction*

[2] modelcontextprotocol, "GitHub - modelcontextprotocol/python-sdk: The official Python SDK for Model Context Protocol servers and clients," GitHub, Mar. 14, 2025. *https://github.com/modelcontextprotocol/python-sdk*

(continue next page)

[3] jlowin, “GitHub - jlowin/fastmcp: The fast, Pythonic way to build MCP servers and clients,” GitHub, Oct. 26, 2025. *https://github.com/jlowin/fastmcp*

[4] langchain-ai, “GitHub - langchain-ai/langchain-mcp-adapters: LangChain MCP,” GitHub, Dec. 09, 2025. *https://github.com/langchain-ai/langchain-mcp-adapters* (accessed Feb. 11, 2026).

Simulation & Environment

[1] “Mesa: Agent-based modeling in Python — Mesa .1 documentation,” Readthedocs.io, 2025. *https://mesa.readthedocs.io/*

Agent Frameworks

[1] crewAI Inc, “Home,” crewai.com. *https://docs.crewai.com/*

[2] “autogen_ext.tools.mcp — AutoGen,” Github.io, 2024. *https://microsoft.github.io/autogen/stable/reference/python/autogen_ext.tools.mcp.html* (accessed Feb. 11, 2026).

Vercel AI SDK

[1] “AI SDK by Vercel,” Ai-sdk.dev, 2025. *https://ai-sdk.dev/docs/introduction*

CASE STUDY 4:

The Inclusive Citizen – Multilingual AI for Public Services

Track: Natural Language Processing (NLP)

Primary Goal: SDG 10: Reduced Inequalities (Target 10.2)

1. Real-World Context

ASEAN is a mosaic of thousands of languages and dialects. However, digital government portals (e-Gov) often use a single dominant national language and complex official terminology. This creates a barrier for elderly users, migrant workers, and rural communities, leading to exclusion from essential public health, legal, and social services.

2. Problem Statement

There is a massive "Information Gap" in digital public services. We need an NLP-powered assistant that does not just translate, but simplifies and summarizes complex official information into multiple local languages and dialects. This ensures that every citizen, regardless of their literacy level or language, can understand and access government support.

3. Technical Challenge & Sub-tasks

- **Dialect-Aware Translation:** Move beyond standard national languages to include local dialects (e.g., regional dialects in Indonesia or the Philippines).
- **Text Simplification (Lexical Level):** Automatically replace complex legal or medical jargon with simple, everyday language (e.g., 5th-grade reading level).
- **Recursive Summarization:** Implement a pipeline that condenses long policy documents into 3–5 actionable bullet points for the user.
- **Cross-Lingual Information Retrieval:** Develop a system where a user can ask a question in a dialect and get an answer sourced from a national-language document.

4. Technical Feasibility & Constraints

- **Accuracy & Hallucination Control:** In public services, misinformation is dangerous. The AI must be grounded in official documents (e.g., using RAG - Retrieval Augmented Generation).
- **Low-Resource Languages:** Participants should demonstrate how they handle languages that don't have massive online datasets.
- **Inclusivity UI:** The interface must be designed for users with low digital literacy (e.g., Voice-to-Text or WhatsApp-style UI).

5. Recommended Data Sources & Toolkits

- **Suggested Frameworks:** LangChain or LlamaIndex (for RAG), Hugging Face Transformers.
- **Suggested Models:** SEA-LION v4 (AI Singapore), Sailor 2 (SEA AI Lab), Typhoon 2.1 (SCB 10X), Llama-Sahabat-AI-v2-70B (Indosat/GoTo), ILMU (YTL AI Labs), and VinaLLaMA (Vietnamese AI).
- **NLP Tools:** NLTK / Spacy

6. Expected Deliverables

- **Multilingual AI Assistant:** A functional web or mobile interface demonstrating translation and simplification.
- **Architecture Diagram:** A map showing how the AI retrieves official information and transforms it for the end-user.

7. Reference Section

Data & Research Hubs

[1] "facebook/nllb-200-distilled-600M · Hugging Face," Huggingface.co, 2022. *https://huggingface.co/facebook/nllb-200-distilled-600M*

[2] SEACrowd, "SEACrowd - Advancing AI for Southeast Asia | SEACrowd," SEACrowd, 2026. *https://seacrowd.org/* (accessed Feb. 11, 2026).

[3] "Southeast Asian Languages in One Network Data(SEALD) - AI Singapore," Mar. 11, 2024. *https://aisingapore.org/aiproducts/southeast-asian-languages-in-one-network-data-seald/*

[4] asrafulsyifaa, "GitHub - asrafulsyifaa/Malay-Dataset: Text corpus for Bahasa Malaysia, *https://malaya.readthedocs.io/en/latest/Dataset.html,*" *GitHub, 2025. https://github.com/asrafulsyifaa/Malay-Dataset* (accessed Feb. 11, 2026).

(continue next page)

Regional AI Models

[1] "SEA-LION (Southeast Asian Languages In One Network)," GitHub, Dec. 14, 2023.
https://github.com/aisingapore/sealion

[2] sail-sg, "GitHub - sail-sg/sailor2: Sailor2: Sailing in South-East Asia with Inclusive Multilingual LLMs," GitHub, 2025.
https://github.com/sail-sg/sailor2
(accessed Feb. 11, 2026).

[3] scb-10x, "GitHub - scb-10x/typhoon2-audio: The repository of Typhoon2-Audio, Thai audio-language model that supports speech-in and speech-out," GitHub, 2025.
https://github.com/scb-10x/typhoon2-audio
(accessed Feb. 11, 2026).

[4] UMxYTL-AI-Labs, "GitHub - UMxYTL-AI-Labs/MalayMMLU: [MalayMMLU] This is the first-ever Bahasa Melayu multitask benchmark designed to elevate the performance of Large Language Models (LLMs) and Large Vision Language Models (LVLMs).," GitHub, 2025.
https://github.com/UMxYTL-AI-Labs/MalayMMLU
(accessed Feb. 11, 2026).

[5] "https://github.com/vilm-ai/vinallama - Google Search," Google.com, 2023.
https://www.google.com/search?q=https://github.com/vilm-ai/vinallama&authuser=2
(accessed Feb. 11, 2026).

(continue next page)

Evaluation Benchmarks

[1] "Multilexnorm++ Achieves 5-Language Asian Lexical Normalization Benchmark For Improved NLP," *Quantum Zeitgeist*, Jan. 28, 2026. *https://quantumzeitgeist.com/nlp-multilexnorm-achieves-language-asian-lexical/* (accessed Feb. 11, 2026).

[2] S. Pawar et al., "Survey of Cultural Awareness in Language Models: Text and Beyond," arXiv.org, 2024. *https://arxiv.org/abs/2411.00860*

• End of case study booklet •